



MCT5210 – Control Engineering IV

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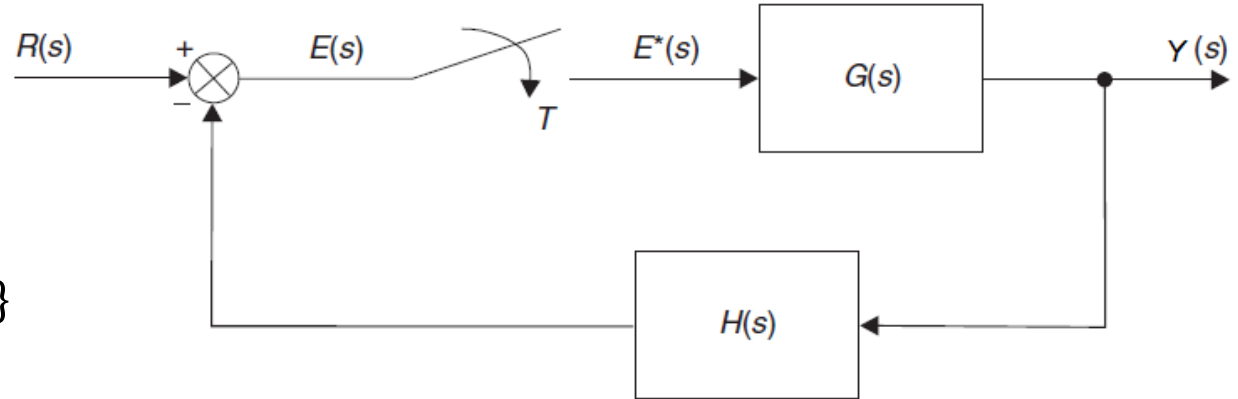
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The Closed-loop Transfer Function

- For an **error only** sampled system, there is no sampler between $G(s)$ and $H(s)$ in the closed-loop and the transfer function is expressed as:

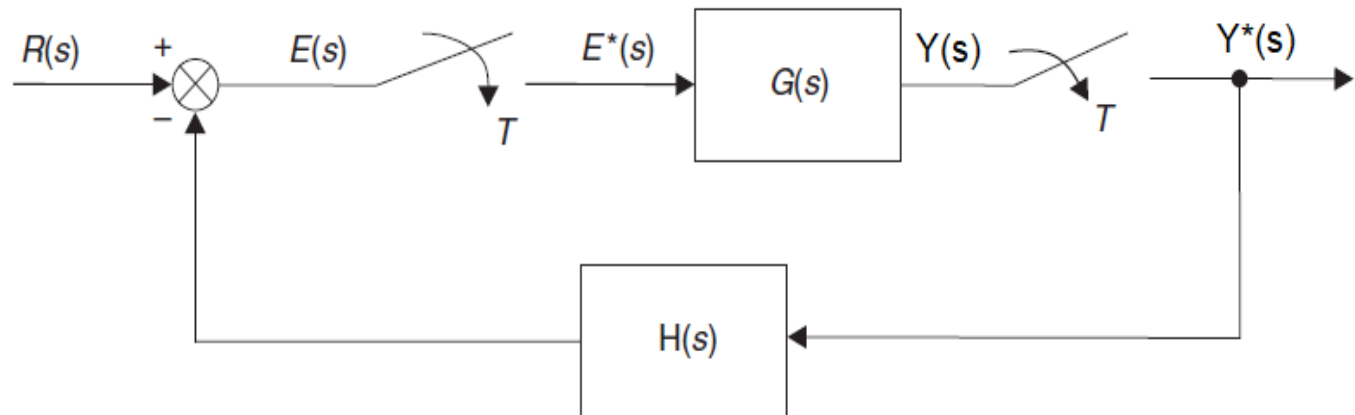
$$\frac{Y(z)}{R(z)} = \frac{G(z)}{1 + GH(z)}$$

where $GH(z) = Z\{GH(s)\}$



- For an **error** and **output** sampled system, there is a sampler between $G(s)$ and $H(s)$ and the transfer function is now written as:

$$\frac{Y(z)}{R(z)} = \frac{G(z)}{1 + G(z)H(z)}$$



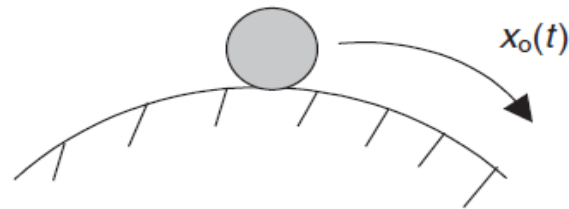


System Stability

- Stability of closed-loop systems is of **utmost** importance in control system design.
- For a **stable** system, a **bounded input** should result in a corresponding **bounded output**.
- Recall that a necessary and sufficient condition for a continuous feedback system to be stable is that all the poles of the system have **negative real** parts.
- That is, the **real** roots of the characteristic equation must lie on the left of the s-plane.
- A system having roots on the $j\omega$ axis is **marginally** stable
- The figure below illustrates the **concept** of stability.



(a) Stable



(b) Unstable

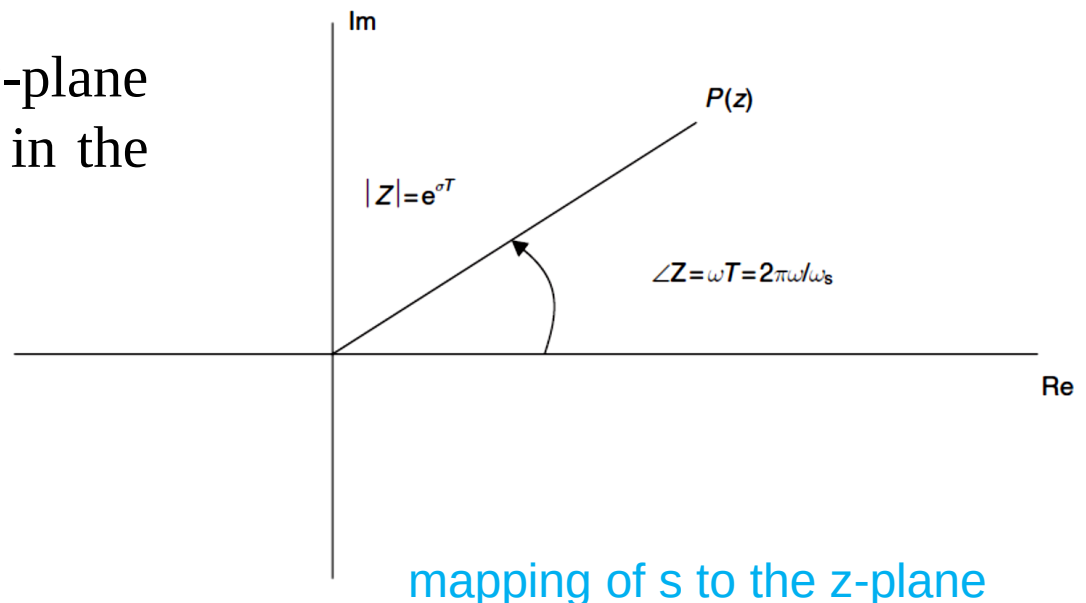
Stability in the z-plane

- Stability analysis of **discrete** systems can be conducted in the **z-plane** just like is done for continuous systems in the s-plane.
- The s-plane can be **mapped** onto the z-plane using the relationship **$z = e^{sT}$** .
- Since $s = \sigma \pm j\omega \Rightarrow z = e^{(\sigma \pm j\omega)T} = e^{\sigma T} e^{j\omega T}$.
- If $e^{\sigma T} = |z|$ and $T = 2\pi/\omega_s$ then z can be expressed as

$$z = |z|e^{j(2\pi\omega/\omega_s)}$$

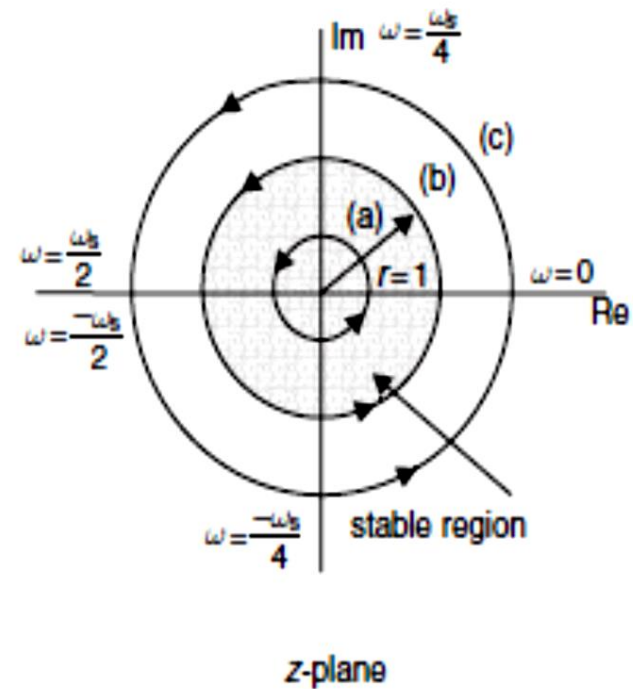
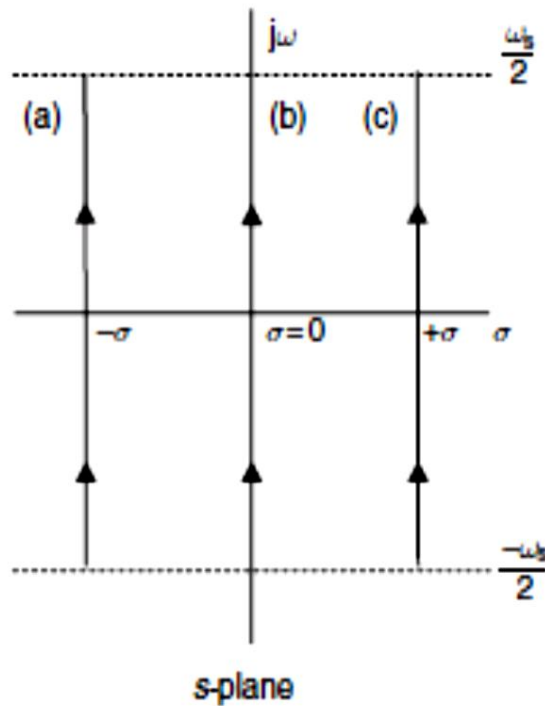
where ω_s is the sampling frequency.

- The mapping the of the s-plane to the z-plane thus results in the polar diagram shown.



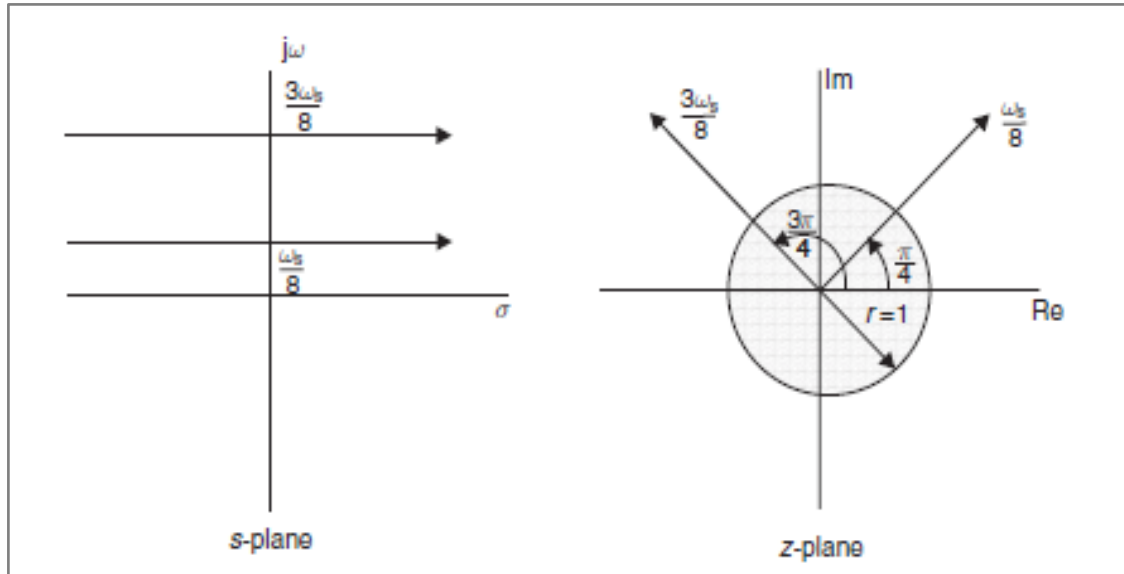
Stability in the z-plane

- The diagram below shows the mapping of constant σ (constant settling time) from the s to the z-plane.
- It can be observed that the **stable** region (left hand side) of the s-plane corresponds to the region within a circle of unit radius (**unit circle**).



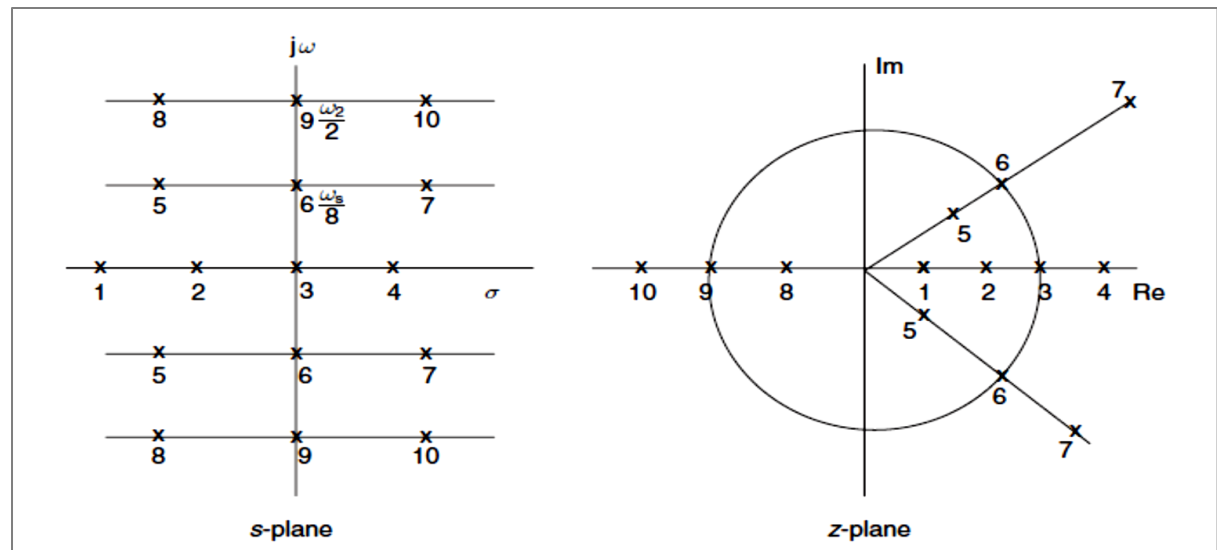
Mapping from the s to the z-plane

Stability in the z-plane



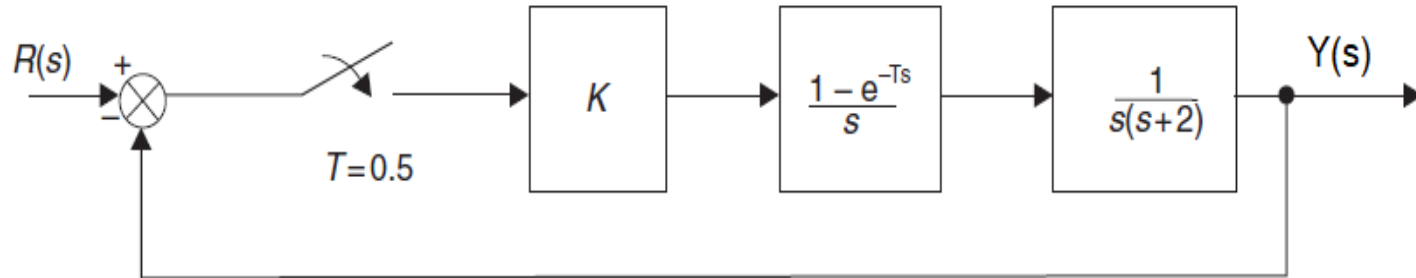
Mapping of constant ω (constant transient frequency) from the s to the z-plane.

Corresponding pole maps from the s to the z-plane



Stability in the z-plane

Example: Consider the system shown below.



If the open loop z transfer function of the plant and ZOH to a step input is

$$\frac{Y(z)}{R(z)} = \frac{0.092z + 0.07}{z^2 - 1.368z + 0.368}$$

- Find the closed loop discrete transfer function with the compensator K .
- Analyse the stability of the system when $K = 1$ and $K = 10$
- Find the steady state error for any stable case in b above.

Stability in the z-plane

a) The closed-loop transfer function is

$$\frac{Y(z)}{R(z)} = \frac{\frac{K(0.092z + 0.07)}{z^2 - 1.368z + 0.368}}{1 + \frac{K(0.092z + 0.07)}{z^2 - 1.368z + 0.368}}$$

$$\frac{Y(z)}{R(z)} = \frac{0.092Kz + 0.07K}{z^2 - (1.368 - 0.092K)z + 0.368 + 0.07K}$$

b) The characteristic equation is $1 + GH(z) = 0$

When $K = 1$, the system is defined as

$$\frac{Y(z)}{R(z)} = \frac{0.092z + 0.07}{z^2 - 1.276z + 0.438}$$

Thus the roots of the characteristic equation are $0.6380 + 0.1759j$ and $0.6380 - 0.1759j$

$|z| = 0.6618$ and hence the system is **stable** since the poles lie **within** the **unit circle**.

Stability in the z-plane

When $K = 10$, the system is defined as

$$\frac{Y(z)}{R(z)} = \frac{0.92z + 0.7}{z^2 - 0.448z + 1.068}$$

Thus the roots of the characteristic equation are $0.2240 + 1.0089j$ and $0.2240 - 1.0089j$

Hence the system is **unstable** since the poles lie **outside** the **unit** circle.

c) The steady state gain is defined from the **final value theorem**

$$y(\infty) = \lim_{z \rightarrow 1} \left[\left(\frac{z-1}{z} \right) \frac{Y(z)}{R(z)} R(z) \right]$$

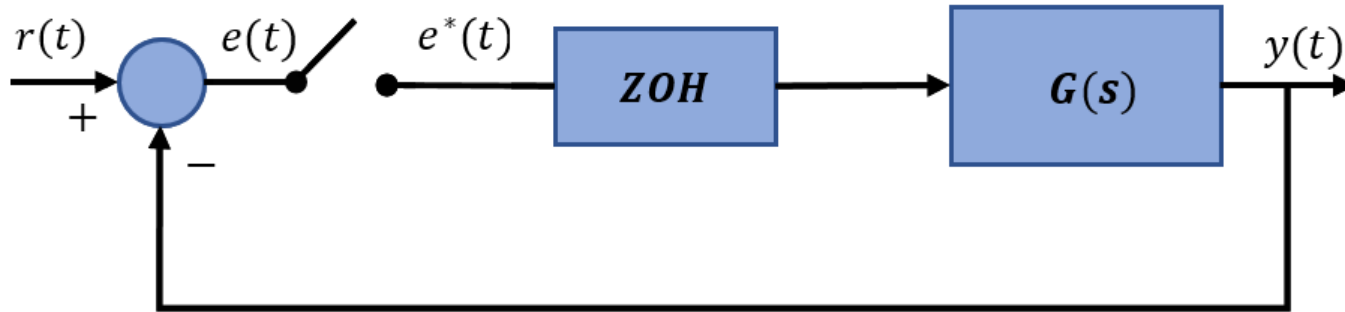
For the stable case ($K=1$) the final value is

$$y(\infty) = \lim_{z \rightarrow 1} \left[\left(\frac{z-1}{z} \right) \left(\frac{0.092z + 0.07}{z^2 - 1.276z + 0.438} \right) \left(\frac{z}{z-1} \right) \right]$$
$$y(\infty) = \frac{0.092 + 0.07}{1 - 1.276 + 0.438} = 1$$

Thus the steady state error = $1 - 1 = 0$

Stability in the z-plane

Example 2: Consider the system shown below.



If the transfer function of the plant is given as

$$G(s) = \frac{1}{s(s+1)}$$

- Find the closed loop discrete transfer function with a sample time of 1 s.
- Analyse the stability of the system.